

Shenglai Zeng

🌐 Personal Website

🎓 Google Scholar

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RESEARCH INTEREST

I am a fourth-year Ph.D. candidate at Michigan State University, advised by Professor Jiliang Tang, and I am on the job market for positions starting in 2027. My research builds **personalized, proactive, and safe agents**. I work primarily on the **agent harness** — the self-improvement, memory, retrieval, and evaluation/environment machinery that determines what an agent can actually do — and extend it to **multimodal interaction** and **post-training**. I have published 25+ papers at NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, KDD, and CVPR, received the Best Paper Award of IEEE Transactions on Cloud Computing (2023) and an ACL-2026 SAC Highlight (top 1%, Best Paper Candidate), and have industry research experience at Meta AI, Amazon, and Baidu.

EDUCATION

Michigan State University

DSE Lab/Ph.D. Candidate in Computer Science and Engineering

East Lansing, U.S

Sept 2023-Present

Expected Graduation: May 2027

Advisor: Jiliang Tang

Lab: Data Science and Engineering Lab

Research Direction: Trustworthy AI, Large language models (LLMs)

University of Electronic Science and Technology of China

Yingcai Honor School/B.Sc in Computer Science and Engineering

Chengdu, China

Sept 2019 - Jun 2023

CGPA: 3.98/4.00

Weighted Average: 93.97/100 (1st among 100 students)

Honors: The Most Outstanding Students Award of UESTC

PREPRINTS

- Xianxuan Long, Zhikai Chen, **Shenglai Zeng**, Shouren Wang, Kai Guo, Jiliang Tang
MemTrace: Probing What Final Accuracy Misses in Long-Term Memory
Pre-print
- Xinnan Dai, Chung-Hsiang Lo, Kai Guo, **Shenglai Zeng**, Dongsheng Luo, Jiliang Tang
Uncovering Graph Reasoning in Decoder-only Transformers with Circuit Tracing
Pre-print
- Zhikai Chen, Jialiang Gu, Junyu Yin, Xianxuan Long, **Shenglai Zeng**, Xiaoze Liu, Kai Guo, Keren Zhou, Jiliang Tang
Exploring Cross-Scenario Generality of Agentic Memory Systems: Diagnostics and a Strong Baseline
Submitted to ACL ARR
- Xinnan Dai, Kai Guo, Chung-Hsiang Lo, **Shenglai Zeng**, Jiayuan Ding, Dongsheng Luo, Subhabrata Mukherjee, Jiliang Tang
GraphGhosts: Tracing Reasoning Structures Behind Large Language Models
Submitted to ACL ARR
- Yuping Lin, Pengfei He, Haoran Zhao, **Shenglai Zeng**, Zhen Xiang, Hui Liu, Jiliang Tang
Structural Alignment Faking: Hiding Malicious Capabilities in Mixture-of-Experts LLMs
Pre-print
- Jie Ren, Han Xu, Pengfei He, Yingqian Cui, **Shenglai Zeng**, Jiankun Zhang, Hongzhi Wen, Jiayuan

PUBLICATIONS

IMPACT: 1340+ Citations

- Yuetai Li, Yichen Feng, Zhangchen Xu, Zixian Ma, Kaiyuan Zheng, Fengqing Jiang, Xinghua Sun, Rulin Shao, Zichen Chen, Yue Huang, Xinyang Han, Brian Lee, Kayla Xu, **Shenglai Zeng**, Hang Hua, Xiangliang Zhang, Basel Alomair, Ranjay Krishna, Luke Zettlemoyer, Pang Wei Koh, Bhaskar Ramasubramanian, Luyao Niu, Xiang Yue, Radha Poovendran
JobBench: Aligning Agent Work With Human Will
NeurIPS-2026 Datasets & Benchmarks Track
- **Shenglai Zeng**, Qirui Wang, Kai Guo, Xinnan Dai, Xianxuan Long, Hui Liu
Magnifying What Matters: Attention-Guided Adaptive Rendering for Visual Text Comprehension
NeurIPS-2026
- Kai Guo, Xinnan Dai, **Shenglai Zeng**, Harry Shomer, Haoyu Han, Yu Wang, Jiliang Tang
Beyond Static Retrieval: Opportunities and Pitfalls of Iterative Retrieval in GraphRAG
EMNLP-2026
- **Shenglai Zeng***, Jiankun Zhang*, Kai Guo, Xinnan Dai, Hui Liu, Jiliang Tang, Yi Chang
Fix Before Search: Benchmarking Agentic Query Visual Pre-processing in Multimodal Retrieval-augmented Generation
ICML-2026
- **Shenglai Zeng**, Tianqi Zheng, Chuan Tian, Dante Everaert, Yau-Shian Wang, Yupin Huang, Michael J. Morais, Rohit Patki, Jinjin Tian, Xinnan Dai, Kai Guo, Monica Xiao Cheng, Hui Liu
Attn-GS: Attention-Guided Context Compression for Efficient Personalized LLMs
ACL-2026(Oral), SAC highlight(1%), Best Paper Candidate
- Xinnan Dai, Kai Yang, Cheng Luo, **Shenglai Zeng**, Kai Guo, Jiliang Tang
When Do Hallucinations Arise? A Graph Perspective on the Evolution of Path Reuse and Path Compression
ICML-2026
- Kai Guo, Xinnan Dai, Zhibo Zhang, Nuohan Lin, **Shenglai Zeng**, Jie Ren, Haoyu Han, Jiliang Tang
Why Retrieval-Augmented Generation Fails: A Graph Perspective
KDD-2026
- Jie Ren, Zhenwei Dai, Xianfeng Tang, Yue Xing, **Shenglai Zeng**, Hui Liu, Jingying Zeng, Qiankun Peng, Samarth Varshney, Suhang Wang, Qi He, Charu C. Aggarwal, Hui Liu
Keeping an Eye on LLM Unlearning: The Hidden Risk and Remedy
NeurIPS-2025
- **Shenglai Zeng**, Jiankun Zhang, Pengfei He, Jie Ren, Tianqi Zheng, Hanqing Lu, Han Xu, Hui Liu, Yue Xing, Jiliang Tang
Mitigating the privacy issues in retrieval-augmented generation (rag) via pure synthetic data
EMNLP-2025-main
- Jiankun Zhang*, **Shenglai Zeng***, Jie Ren, Tianqi Zheng, Hui Liu, Xianfeng Tang, Yi Chang
Beyond Text: Unveiling Privacy Vulnerabilities in Multi-modal Retrieval-Augmented Generation
EMNLP-2025-main
- Kai Guo, Harry Shomer, **Shenglai Zeng**, Haoyu Han, Yu Wang, Jiliang Tang
Empowering GraphRAG with Knowledge Filtering and Integration
EMNLP-2025-main
- **Shenglai Zeng**, Pengfei He, Kai Guo, Tianqi Zheng, Hanqing Lu, Yue Xing, Hui Liu
Towards Context-Robust LLMs: A Gated Representation Fine-tuning Approach

ACL-2025-main

- Bo Wang, Weiyi He, Pengfei He, **Shenglai Zeng**, Zhen Xiang, Yue Xing, Jiliang Tang
Unveiling privacy risks in llm agent memory
ACL-2025-main
- Bingheng Li, Zhikai Chen, Haoyu Han, **Shenglai Zeng**, Jingzhe Liu, Jiliang Tang
Unveiling Mode Connectivity in Graph Neural Networks
KDD-2025
- **Shenglai Zeng**, Jiankun Zhang, Bingheng Li, Yuping Lin, Tianqi Zheng, Dante Everaert, Hanqing Lu, Hui Liu, Yue Xing, Monica Xiao Cheng, Jiliang Tang
Towards Knowledge Checking in Retrieval-augmented Generation: A Representation Perspective
NAACL-2025-main(Oral)
- Jie Ren, Kangrui Chen, Yingqian Cui, **Shenglai Zeng**, Hui Liu, Yue Xing, Jiliang Tang, Lingjuan Lyu
Six-cd: Benchmarking concept removals for benign text-to-image diffusion models
CVPR-2025
- Pengfei He, Yue Xing, Han Xu, Jie Ren, Yingqian Cui, **Shenglai Zeng**, Jiliang Tang, Makoto Yamada, Mohammad Sabokrou
Stealthy Backdoor Attack via Confidence-driven Sampling
TMLR
- Jie Ren, Yaxin Li, **Shenglai Zeng**, Han Xu, Lingjuan Lyu, Yue Xing, Jiliang Tang
Unveiling and mitigating memorization in text-to-image diffusion models through cross attention
ECCV-2024
- Han Xu, Jie Ren, Pengfei He, Yingqian Cui, **Shenglai Zeng**, Hui Liu, Jiliang Tang, Amy Liu
On the Generalization of Training-based ChatGPT Detection Methods
EMNLP-2024-Findings
- **Shenglai Zeng***, Yaxin Li*, Jie Ren, Yiding Liu, Han Xu, Pengfei He, Yue Xing, Shuaiqiang Wang, Jiliang Tang, Dawei Yin
Exploring Memorization in Fine-tuned Language Models
ACL-2024
- **Shenglai Zeng***, Jiankun Zhang*, Pengfei He, Yue Xing, Yiding Liu, Han Xu, Jie Ren, Shuaiqiang Wang, Dawei Yin, Yi Chang, Jiliang Tang
The Good and The Bad: Exploring Privacy Issues in Retrieval-Augmented Generation (RAG)
ACL-2024-findings
- Pengfei He, Han Xu, Jie Ren, Yingqian Cui, **Shenglai Zeng**, Hui Liu, Charu Aggarwal, Jiliang Tang
Sharpness-aware Data Poisoning Attack
ICLR-2024 Spotlight
- Juanhui Li, Harry Shomer, Haitao Mao, **Shenglai Zeng**, Yao Ma, Neil Shah, Jiliang Tang, Dawei Yin
Evaluating graph neural networks for link prediction: Current pitfalls and new benchmarking
NeurIPS-2023 Datasets & Benchmarks Track
- **Shenglai Zeng**, Zonghang Li, Hongfang Yu, Zhihao Zhang, Long Luo, Bo Li, Dusit Niyato
HFedMS: Heterogeneous Federated Learning with Memorable Data Semantics in Industrial Meta-universe
IEEE Transactions on Cloud Computing, 2023 Best Paper
- **Shenglai Zeng**, Zonghang Li, Hongfang Yu, Yihong He, Zenglin Xu, Dusit Niyato, Han Yu
Heterogeneous Federated Learning via Grouped Sequential-to-Parallel Training
International Conference on Database Systems for Advanced Applications (DASFAA-2022)
- Jiaqi Wang*, **Shenglai Zeng***, Zewei Long, Yaqing Wang, Houping Xiao, Fenglong Ma
Knowledge-Enhanced Semi-Supervised Federated Learning for Aggregating Heterogeneous Lightweight Clients in IoT

Patent

- **Shenglai Zeng**, Zonghang Li, Yihong He, Xun Zhang, Hongfang Yu, Gang Sun
"A Hierarchical User Training Management Architecture and Training Strategy for Non-i.i.d Data" Chinese patent

RESEARCH EXPERIENCE

DSE Lab, Michigan State University

Research Assistant/Research on Trustworthy AI and LLM-safety

Lansing, U.S

Sept 2023 - Present

-Advisor: Professor Jiliang Tang

- Integrate IR with LLMs/MLLMs (RAG/MRAG)
- LLM Personalization (Personalization)
- Agent Harness/Memory (Agentic AI)
- Develop a deeper understanding of the underlying mechanisms behind LLMs, and leverage these insights to guide data generation (Data Centric AI& Mechanical understanding)
- Trustworthy AI (Privacy, Factuality, Grounding)

Meta AI

Research Scientist Intern/Research on RAG/Agentic AI

Seattle, USA

Summer 2026

-Mentors: Yang Xiao, Luna Xin Dong, Scott Wen-tau Yih

- Collaborate with cross-organizational researchers spanning Meta Superintelligence Labs, FAIR, and Reality Labs on next-generation proactive multimodal agentic AI.

Search Science Team, Amazon

Applied Scientist Intern/Research on Retrieval-augmented Generation and personalization

CA, USA

Summer 2024, 2025

-Mentor: Tianqi Zheng, Dante Everaert

-Manager: Hanqing Lu, Chuan Tian, Monica Xiao Cheng

- Representation finetuning for robust RAG [ACL-2025].
- Utilize LLMs' internal behavior to conduct knowledge checking in RAG [NAACL-2025(oral)].
- Privacy of multi-modality RAG [EMNLP-2025]
- Synthetic data for RAG [EMNLP-2025]

Search Science Team, Baidu.Inc

Research Intern/Research on the memorization of LLM

Beijing, China

May 2023 - May 2024

-Mentor: Dr. Yiding Liu and Dr. Dawei Yin

- Investigating the memorization behavior and privacy implications of fine-tuned LLMs[ACL-2024].
- Currently worked on privacy risks of Retrieval LMs and AI-agents[ACL-2024].

AWARDS AND ACHIEVEMENTS

- SAC Highlights(Top 1%), Best Paper Candidates, ACL-2026
- Best Paper Award, IEEE Transactions on Cloud Computing (JCR Q1)
- The Most Outstanding Students Award of UESTC (Highest honor in UESTC, Only 10 students are awarded)
- National Scholarship in the session of 2019-2020.(Highest honor of undergraduate student)
- WAC Scholarship in the session of 2020-2021.(Only 10 undergraduate students in UESTC are awarded each year)
- 1st Outstanding Academic Scholarship in 2020,2021,and 2022.(Top 5 % students)

Services

- **Conference Reviewer / Program Committee:** NeurIPS, ICML, ICLR; ACL, EMNLP, NAACL (via ACL Rolling Review); MICCAI Workshop on Distributed, Collaborative and Federated Learning (DeCaF-2022)
- **Journal Reviewer:** ACM Transactions on Information Systems (TOIS), IEEE TKDE, IEEE TMC, IEEE TPDS, IEEE Transactions on Information Forensics & Security, IEEE Transactions on Services Computing, IEEE Transactions on Vehicular Technology, IEEE Internet of Things Journal
- Reviewed **30+** papers.